

ShopKart E-Commerce Business Analytics

From Database Design to Strategic Business Intelligence

Submitted by: Chetan Prajapat

File: ACE_Project_ChetanPrajapat

Course: BCA 4th Sem BCA

College: Sage university indore

Enrolment no: 24COA2BCA0044

GitHub: https://github.com/grchetan/ACE_Project_Query_ChetanPrajapat/

1. Introduction — Meet ShopKart

ShopKart is an online marketplace operating across major Indian cities, selling products across three core categories. This case study demonstrates a complete SQL Data Analyst workflow — from database design and data insertion to multi-level business intelligence queries.

1.1 Product Categories

Category	Products	Price Range
Electronics	Laptops, Mobiles, Headphones	High-value tech products
Fashion	T-Shirts, Jeans	High volume, lower price-point
Home Appliances	Microwave, Refrigerator, Air Conditioner	Large-ticket, strong margins

1.2 The Management Problem

Despite growing order volumes, ShopKart's leadership faces five critical gaps that require data-driven answers:

- **Profit Clarity** — Sales are increasing, but profitability remains unclear. Management cannot distinguish high-revenue products from high-profit ones.
- **City-Level Anomalies** — Certain cities show high order counts but surprisingly low revenue.
- **Top Customer Identification** — The business needs to identify its most valuable customers for targeted retention strategies.
- **Category Performance** — No clear view of which product categories are driving profit versus just volume.
- **Monthly Growth Tracking** — A structured monthly order trend report is needed to understand growth trajectory.

2. Step 1 — Create the Database

The first step is to initialize the ShopKart database environment in MySQL. All subsequent tables and data will live within this database.

```
CREATE DATABASE shopkart_db;  
USE shopkart_db;
```

This establishes the working schema. The USE statement ensures all subsequent DDL and DML commands are executed within the correct database context — a critical step before any table creation or data insertion.

3. Step 2 — Database Schema Design

ShopKart's relational database is built on five interconnected tables. Each table has a clearly defined primary key, and foreign keys enforce referential integrity across the schema.

3.1 Schema Overview

Table	Primary Key	Foreign Key(s)	Purpose
customers	customer_id	—	Customer profiles and demographics
categories	category_id	—	Product category hierarchy
products	product_id	category_id → categories	Product catalog with price and cost
orders	order_id	customer_id → customers	Order header with status and city
order_items	order_item_id	order_id → orders, product_id → products	Line items linking orders and products

The schema follows a classic star-like structure: Orders sits at the center, linked to Customers (who placed the order), Order Items (what was ordered), and Products (which belong to Categories). This design enables powerful multi-table JOIN queries for business analysis.

3.2 Table Creation — DDL

Customers Table

```
CREATE TABLE customers (  
  customer_id INT PRIMARY KEY,  
  customer_name VARCHAR(100),  
  gender VARCHAR(10),  
  city VARCHAR(50),  
  signup_date DATE  
);
```

Categories Table

```
CREATE TABLE categories (  
  category_id INT PRIMARY KEY,  
  category_name VARCHAR(50)  
);
```

Products Table

```
CREATE TABLE products (  
  product_id INT PRIMARY KEY,  
  product_name VARCHAR(100),  
  category_id INT,  
  price DECIMAL(10,2),  
  cost DECIMAL(10,2),  
  FOREIGN KEY (category_id) REFERENCES categories(category_id)  
);
```

Orders Table

```
CREATE TABLE orders (  
    order_id INT PRIMARY KEY,  
    customer_id INT,  
    order_date DATE,  
    city VARCHAR(50),  
    order_status VARCHAR(20),  
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id)  
);
```

Order Items Table

```
CREATE TABLE order_items (  
    order_item_id INT PRIMARY KEY,  
    order_id INT,  
    product_id INT,  
    quantity INT,  
    FOREIGN KEY (order_id) REFERENCES orders(order_id),  
    FOREIGN KEY (product_id) REFERENCES products(product_id)  
);
```

4. Step 3 — Insert Dataset

The following INSERT statements populate all five tables with ShopKart's sample dataset.

4.1 Customers Data

```
INSERT INTO customers VALUES  
(1, 'Rahul Sharma', 'Male', 'Delhi', '2022-01-10'),  
(2, 'Priya Mehta', 'Female', 'Mumbai', '2021-05-20'),  
(3, 'Amit Patel', 'Male', 'Ahmedabad', '2023-02-14'),  
(4, 'Sneha Reddy', 'Female', 'Hyderabad', '2022-11-01'),  
(5, 'Karan Verma', 'Male', 'Delhi', '2020-08-18'),  
(6, 'Neha Nair', 'Female', 'Chennai', '2021-09-09'),  
(7, 'Rohit Gupta', 'Male', 'Pune', '2023-03-12'),  
(8, 'Anjali Singh', 'Female', 'Bangalore', '2022-06-25');
```

Output:

Result Grid		Filter Rows:		Edit:	Export/1
	customer_id	customer_name	gender	city	signup_date
▶	1	Rahul Sharma	Male	Delhi	2022-01-10
	2	Priya Mehta	Female	Mumbai	2021-05-20
	3	Amit Patel	Male	Ahmedabad	2023-02-14
	4	Sneha Reddy	Female	Hyderabad	2022-11-01
	5	Karan Verma	Male	Delhi	2020-08-18
	6	Neha Nair	Female	Chennai	2021-09-09
	7	Rohit Gupta	Male	Pune	2023-03-12
	8	Anjali Singh	Female	Bangalore	2022-06-25
*	NULL	NULL	NULL	NULL	NULL

4.2 Categories Data

```
INSERT INTO categories VALUES  
(1, 'Electronics'),  
(2, 'Fashion'),  
(3, 'Home Appliances');
```

Output:

Result Grid	Filter Rows:	Edit:
category_id	category_name	
1	Electronics	
2	Fashion	
3	Home Appliances	
NULL	NULL	

4.3 Products Data

```
INSERT INTO products VALUES
(101, 'Laptop', 1, 80000, 65000),
(102, 'Mobile', 1, 30000, 22000),
(103, 'Headphones', 1, 2000, 1200),
(104, 'T-Shirt', 2, 800, 300),
(105, 'Jeans', 2, 2000, 900),
(106, 'Microwave', 3, 7000, 5000),
(107, 'Refrigerator', 3, 30000, 24000),
(108, 'Air Conditioner', 3, 45000, 35000);
```

Output:

Result Grid

Filter Rows:

Edit:

Export/Import:

	product_id	product_name	category_id	price	cost
▶	101	Laptop	1	80000.00	65000.00
	102	Mobile	1	30000.00	22000.00
	103	Headphones	1	2000.00	1200.00
	104	T-Shirt	2	800.00	300.00
	105	Jeans	2	2000.00	900.00
	106	Microwave	3	7000.00	5000.00
	107	Refrigerator	3	30000.00	24000.00
	108	Air Conditioner	3	45000.00	35000.00
✱	NULL	NULL	NULL	NULL	NULL

4.4 Orders Data

```
INSERT INTO orders VALUES
(1001, 1, '2024-01-10', 'Delhi', 'Delivered'),
(1002, 2, '2024-01-11', 'Mumbai', 'Delivered'),
(1003, 3, '2024-01-12', 'Ahmedabad', 'Cancelled'),
(1004, 4, '2024-01-13', 'Hyderabad', 'Delivered'),
(1005, 5, '2024-01-15', 'Delhi', 'Delivered'),
(1006, 6, '2024-01-16', 'Chennai', 'Pending'),
(1007, 1, '2024-02-02', 'Delhi', 'Delivered'),
(1008, 7, '2024-02-05', 'Pune', 'Delivered'),
(1009, 8, '2024-02-08', 'Bangalore', 'Delivered'),
(1010, 2, '2024-02-10', 'Mumbai', 'Cancelled');
```

Output:

Result Grid					
Filter Rows:					
	order_id	customer_id	order_date	city	order_status
▶	1001	1	2024-01-10	Delhi	Delivered
	1002	2	2024-01-11	Mumbai	Delivered
	1003	3	2024-01-12	Ahmedabad	Cancelled
	1004	4	2024-01-13	Hyderabad	Delivered
	1005	5	2024-01-15	Delhi	Delivered
	1006	6	2024-01-16	Chennai	Pending
	1007	1	2024-02-02	Delhi	Delivered
	1008	7	2024-02-05	Pune	Delivered
	1009	8	2024-02-08	Bangalore	Delivered
	1010	2	2024-02-10	Mumbai	Cancelled
•	NULL	NULL	NULL	NULL	NULL

4.5 Order Items Data

```
INSERT INTO order_items VALUES
(1, 1001, 101, 1), -- Order 1001: Laptop x1
(2, 1001, 103, 2), -- Order 1001: Headphones x2
(3, 1002, 102, 1), -- Order 1002: Mobile x1
(4, 1003, 104, 3), -- Order 1003: T-Shirt x3 (Cancelled)
(5, 1004, 106, 1), -- Order 1004: Microwave x1
(6, 1005, 105, 2), -- Order 1005: Jeans x2
(7, 1005, 104, 1), -- Order 1005: T-Shirt x1
(8, 1006, 102, 1), -- Order 1006: Mobile x1 (Pending)
(9, 1007, 108, 1), -- Order 1007: Air Conditioner x1
(10, 1008, 103, 3), -- Order 1008: Headphones x3
(11, 1009, 107, 1), -- Order 1009: Refrigerator x1
(12, 1010, 104, 2); -- Order 1010: T-Shirt x2 (Cancelled)
```

Output:

	order_item_id	order_id	product_id	quantity
▶	1	1001	101	1
	2	1001	103	2
	3	1002	102	1
	4	1003	104	3
	5	1004	106	1
	6	1005	105	2
	7	1005	104	1
	8	1006	102	1
	9	1007	108	1
	10	1008	103	3
	11	1009	107	1
	12	1010	104	2
•	NULL	NULL	NULL	NULL

order_items 6 ×

5. Level 1 — Basic Exploration Queries

Level 1 establishes foundational SQL skills — filtering, sorting, and limiting results. These queries form the bedrock of any data investigation.

Q1 — Customers from Delhi

Returns all customers whose city is Delhi. Uses the WHERE clause to filter rows by city value.

```
SELECT *  
FROM customers  
WHERE city = 'Delhi';
```

Result:

	customer_id	customer_name	gender	city	signup_date
▶	1	Rahul Sharma	Male	Delhi	2022-01-10
	5	Karan Verma	Male	Delhi	2020-08-18
✱	NULL	NULL	NULL	NULL	NULL

Q2 — Latest 5 Orders

Sorts all orders by order_date in descending order and returns only the 5 most recent. Useful for operational dashboards.

```
SELECT *  
FROM orders  
ORDER BY order_date DESC  
LIMIT 5;
```

Result:

	order_id	customer_id	order_date	city	order_status
▶	1010	2	2024-02-10	Mumbai	Cancelled
	1009	8	2024-02-08	Bangalore	Delivered
	1008	7	2024-02-05	Pune	Delivered
	1007	1	2024-02-02	Delhi	Delivered
	1006	6	2024-01-16	Chennai	Pending
✱	NULL	NULL	NULL	NULL	NULL

Q3 — Top 3 Most Expensive Products

Returns the three highest-priced items in the catalog using ORDER BY price DESC and LIMIT 3.

```
SELECT product_name, price  
FROM products  
ORDER BY price DESC  
LIMIT 3;
```

Result:

	product_name	price
▶	Laptop	80000.00
	Air Conditioner	45000.00
	Mobile	30000.00

6. Level 2 — Aggregation Queries

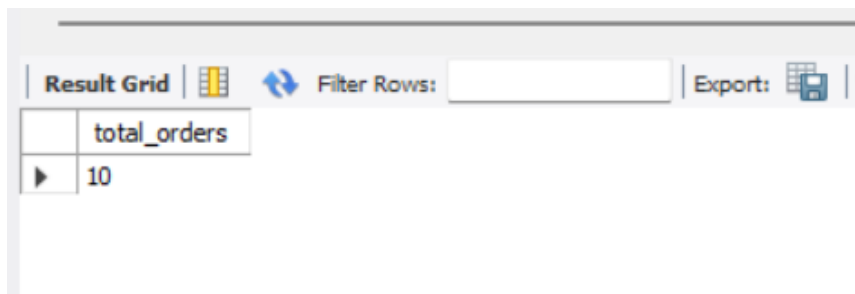
Level 2 introduces aggregate functions and grouping — the core of business reporting. These queries answer 'how many' and 'how much' questions at scale.

Q4 — Total Number of Orders

Uses COUNT() to return the total order volume. This is the simplest aggregation and a baseline KPI for any e-commerce business.

```
SELECT COUNT(*) AS total_orders
FROM orders;
```

Result:



The screenshot shows a database query result interface. At the top, there is a toolbar with a 'Result Grid' button, a 'Filter Rows' input field, and an 'Export' button. Below the toolbar, the query result is displayed in a table with two columns: 'total_orders' and a value of 10.

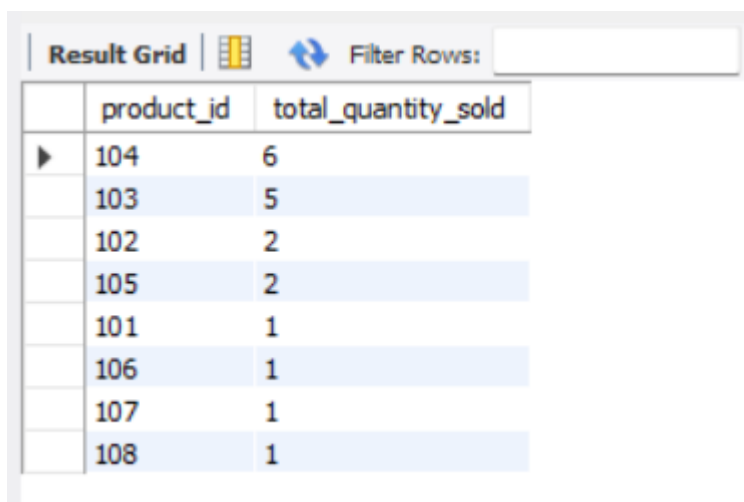
	total_orders
▶	10

Q5 — Total Quantity Sold Per Product

Groups order line items by product and sums quantities. Reveals which products move the most units — critical for inventory planning.

```
SELECT product_id,
SUM(quantity) AS total_quantity_sold
FROM order_items
GROUP BY product_id
ORDER BY total_quantity_sold DESC;
```

Result:



The screenshot shows a database query result interface. At the top, there is a toolbar with a 'Result Grid' button, a 'Filter Rows' input field, and an 'Export' button. Below the toolbar, the query result is displayed in a table with two columns: 'product_id' and 'total_quantity_sold'. The table contains 8 rows of data, sorted by total_quantity_sold in descending order.

	product_id	total_quantity_sold
▶	104	6
	103	5
	102	2
	105	2
	101	1
	106	1
	107	1
	108	1

Note: product_id 102 (Mobile) appears in both a delivered (1002) and a pending order (1006).

Q6 — Cities with More Than 2 Orders

Uses HAVING (not WHERE) to filter after aggregation. HAVING is used when filtering on aggregate functions like COUNT().

```
SELECT city, COUNT(*) AS order_count
FROM orders
GROUP BY city
HAVING COUNT(*) > 2;
```

Result:

city	order_count
Delhi	3

7. Level 3 — Join-Based Real Analysis

Level 3 unlocks the true power of relational databases. By joining multiple tables, we can answer complex business questions that no single table can answer alone.

Q7 — Customer Name with Order ID

Joins customers and orders tables on the shared customer_id key. Produces a human-readable order list with customer names instead of raw IDs.

```
SELECT c.customer_name, o.order_id, o.order_date, o.city, o.order_status
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
ORDER BY o.order_date;
```

Result:

customer_name	order_id	order_date	city	order_status
Rahul Sharma	1001	2024-01-10	Delhi	Delivered
Priya Mehta	1002	2024-01-11	Mumbai	Delivered
Amit Patel	1003	2024-01-12	Ahmedabad	Cancelled
Sneha Reddy	1004	2024-01-13	Hyderabad	Delivered
Karan Verma	1005	2024-01-15	Delhi	Delivered
Neha Nair	1006	2024-01-16	Chennai	Pending
Rahul Sharma	1007	2024-02-02	Delhi	Delivered
Rohit Gupta	1008	2024-02-05	Pune	Delivered
Anjali Singh	1009	2024-02-08	Bangalore	Delivered
Priya Mehta	1010	2024-02-10	Mumbai	Cancelled

Q8 — Product Name in Each Order

A three-table JOIN spanning orders, order_items, and products. This is the most common pattern in e-commerce analytics — the order receipt view.

```
SELECT o.order_id, p.product_name, oi.quantity
```

```
FROM orders o
JOIN order_items oi ON o.order_id = oi.order_id
JOIN products p ON oi.product_id = p.product_id
ORDER BY o.order_id;
```

Result:

order_id	product_name	quantity
1001	Laptop	1
1001	Headphones	2
1002	Mobile	1
1003	T-Shirt	3
1004	Microwave	1
1005	T-Shirt	1
1005	Jeans	2
1006	Mobile	1
1007	Air Conditioner	1
1008	Headphones	3
1009	Refrigerator	1
1010	T-Shirt	2

Q9 — Total Quantity Per Customer

Combines JOIN with GROUP BY to aggregate total purchase quantity per customer — a foundation for customer ranking and segmentation.

```
SELECT c.customer_name, SUM(oi.quantity) AS total_quantity
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY c.customer_name
ORDER BY total_quantity DESC;
```

Result:

customer_name	total_quantity
Rahul Sharma	4
Priya Mehta	3
Amit Patel	3
Karan Verma	3
Rohit Gupta	3
Sneha Reddy	1
Neha Nair	1
Anjali Singh	1

8. Level 4 — Profit Analytics

Profit analytics is one of the most industry-critical SQL skills. The key formula: Profit = (Price - Cost) x Quantity. These queries directly address management's core concern about unclear profitability.

Q10 — Profit Per Product (Per Unit)

Calculates per-unit margin for every product. This reveals that Laptop has the highest absolute margin per unit at Rs. 15,000, while T-Shirt has only Rs. 500 margin per unit.

```
SELECT product_name, price, cost,
       (price - cost) AS profit_per_unit
FROM products
ORDER BY profit_per_unit DESC;
```

Result:

product_name	price	cost	profit_per_unit
Laptop	80000.00	65000.00	15000.00
Air Conditioner	45000.00	35000.00	10000.00
Mobile	30000.00	22000.00	8000.00
Refrigerator	30000.00	24000.00	6000.00
Microwave	7000.00	5000.00	2000.00
Jeans	2000.00	900.00	1100.00
Headphones	2000.00	1200.00	800.00
T-Shirt	800.00	300.00	500.00

Q11 — Category-Wise Total Profit

A four-table JOIN that rolls up profit to the category level. This answers the management question: which category is most profitable — Electronics, Fashion, or Home Appliances?

```
SELECT cat.category_name,
       SUM((p.price - p.cost) * oi.quantity) AS total_profit
FROM categories cat
JOIN products p      ON cat.category_id = p.category_id
JOIN order_items oi  ON p.product_id    = oi.product_id
JOIN orders o        ON oi.order_id     = o.order_id
GROUP BY cat.category_name
ORDER BY total_profit DESC;
```

Result:

category_name	total_profit
Electronics	35000.00
Home Appliances	18000.00
Fashion	5200.00

9. Level 5 — Customer Intelligence

Level 5 moves from transactional reporting to customer behavioral analysis — identifying repeat buyers and high-value purchasers using advanced SQL techniques including subqueries.

Q12 — Customers with More Than 1 Order (Repeat Buyers)

Uses HAVING COUNT() after grouping to isolate repeat customers. Repeat customer rate is a key retention KPI for any e-commerce business.

```
SELECT c.customer_name, COUNT(o.order_id) AS total_orders
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
GROUP BY c.customer_name
HAVING COUNT(o.order_id) > 1;
```

Result:

customer_name	total_orders
Rahul Sharma	2
Priya Mehta	2

Q13 — Customers Above Average Purchase Quantity

Introduces a subquery to dynamically compute the average purchase quantity and use it as a filter threshold. This is a hallmark of intermediate-to-advanced SQL — the inner query calculates the benchmark, the outer query applies it.

```
SELECT c.customer_name, SUM(oi.quantity) AS total_quantity
FROM customers c
JOIN orders o ON c.customer_id = o.customer_id
JOIN order_items oi ON o.order_id = oi.order_id
GROUP BY c.customer_name
HAVING SUM(oi.quantity) > (
    SELECT AVG(total_qty)
    FROM (
        SELECT SUM(quantity) AS total_qty
        FROM order_items
        GROUP BY order_id
    ) subquery
);
```

customer_name	total_quantity
Rahul Sharma	4
Priya Mehta	3
Amit Patel	3
Karan Verma	3
Rohit Gupta	3

The inner subquery computes order-level quantity averages, and the outer query returns only customers whose total purchase volume exceeds that average — enabling dynamic threshold-based filtering.

10. Level 6 — Strategic Business Questions

Level 6 delivers the executive-level insights that directly inform business strategy — city revenue rankings, product profitability, and growth trends over time.

Q14 — City with Highest Revenue

Addresses the management concern about cities with high orders but low revenue. Multiplies price x quantity per city and ranks them to expose geographic revenue distribution.

```
SELECT o.city,  
       SUM(p.price * oi.quantity) AS total_revenue  
FROM orders o  
JOIN order_items oi ON o.order_id = oi.order_id  
JOIN products p ON oi.product_id = p.product_id  
GROUP BY o.city  
ORDER BY total_revenue DESC;
```

Result:

city	total_revenue
Delhi	133800.00
Mumbai	31600.00
Chennai	30000.00
Bangalore	30000.00
Hyderabad	7000.00
Pune	6000.00
Ahmedabad	2400.00

Q15 — Most Profitable Product

Combines margin calculation with quantity sold to find the product generating the most absolute profit — not just the highest price or highest volume.

```
SELECT p.product_name,  
       SUM((p.price - p.cost) * oi.quantity) AS total_profit  
FROM products p  
JOIN order_items oi ON p.product_id = oi.product_id  
GROUP BY p.product_name  
ORDER BY total_profit DESC  
LIMIT 1;
```

Result:

product_name	total_profit
Mobile	16000.00

Q16 — Monthly Order Trend (Jan vs Feb 2024)

Uses MONTH() date function to extract the month from each order date and group by it. Produces the monthly growth report management requested.

```
SELECT MONTH(order_date) AS month_number,  
       MONTHNAME(order_date) AS month_name,  
       COUNT(*) AS total_orders  
FROM orders
```

```
GROUP BY MONTH(order_date), MONTHNAME(order_date)
ORDER BY month_number;
```

Result:

Result Grid	Filter Rows:	Export:	Wrap C
month_number	month_name	total_orders	
1	January	6	
2	February	4	

Business Insight: January had 6 orders and February had 4. While February shows lower volume, trend analysis requires more months of data to determine seasonality vs. decline.

11. Final Capstone — Complete Business Intelligence Report

The manager's final requirement: a comprehensive SQL report that consolidates all analytical dimensions into a single business intelligence deliverable.

Capstone Part A — Top 3 Customers by Total Spend

```
SELECT c.customer_name,
SUM(p.price * oi.quantity) AS total_spend
FROM customers c
JOIN orders o      ON c.customer_id = o.customer_id
JOIN order_items oi ON o.order_id   = oi.order_id
JOIN products p    ON oi.product_id = p.product_id
GROUP BY c.customer_name
ORDER BY total_spend DESC
LIMIT 3;
```

Expected Result:

Result Grid	Filter Rows:	Export:	Wrap Cell
customer_name	total_spend		
Rahul Sharma	129000.00		
Priya Mehta	31600.00		
Neha Nair	30000.00		

Capstone Part B — Revenue Per City

```
SELECT o.city,
COUNT(DISTINCT o.order_id) AS total_orders,
SUM(p.price * oi.quantity) AS total_revenue
FROM orders o
JOIN order_items oi ON o.order_id   = oi.order_id
JOIN products p    ON oi.product_id = p.product_id
GROUP BY o.city
ORDER BY total_revenue DESC;
```

Result Grid				Filter Rows:	Export:	Wrap Cell Content:
	city	total_orders	total_revenue			
▶	Delhi	3	133800.00			
	Mumbai	2	31600.00			
	Bangalore	1	30000.00			
	Chennai	1	30000.00			
	Hyderabad	1	7000.00			
	Pune	1	6000.00			
	Ahmedabad	1	2400.00			

Capstone Part C — Profit Per Category (Full P&L View)

```
SELECT cat.category_name,
       SUM(p.price * oi.quantity) AS total_revenue,
       SUM(p.cost * oi.quantity) AS total_cost,
       SUM((p.price - p.cost) * oi.quantity) AS total_profit,
       ROUND(
         SUM((p.price - p.cost) * oi.quantity) * 100.0 /
         SUM(p.price * oi.quantity), 2
       ) AS profit_margin_pct
FROM categories cat
JOIN products p ON cat.category_id = p.category_id
JOIN order_items oi ON p.product_id = oi.product_id
JOIN orders o ON oi.order_id = o.order_id
GROUP BY cat.category_name
ORDER BY total_profit DESC;
```

Capstone Part D — Repeat Customer Count

```
SELECT COUNT(*) AS repeat_customer_count
FROM (
  SELECT customer_id, COUNT(order_id) AS order_count
  FROM orders
  GROUP BY customer_id
  HAVING COUNT(order_id) > 1
) repeat_customers;
```

Result:

Result Grid		Filter Rows:	Export:	Wrap Cell C
	repeat_customer_count			
▶	2			

Capstone Part E — Cancelled Order Percentage

```
SELECT
  COUNT(*) AS total_orders,
  SUM(CASE WHEN order_status = 'Cancelled' THEN 1 ELSE 0 END) AS cancelled_orders,
  ROUND(
    SUM(CASE WHEN order_status = 'Cancelled' THEN 1 ELSE 0 END) * 100.0
    / COUNT(*), 2
  ) AS cancellation_rate_pct
FROM orders;
```

Result:

	total_orders	cancelled_orders	cancellation_rate_pct
▶	10	2	20.00

12. Project Summary — Skills Progression

This case study covers the full spectrum of SQL skills required for a professional data analyst role, progressing from basic syntax to strategic business intelligence.

Level	Skills Demonstrated	Queries
Level 1 — Basic Exploration	WHERE, ORDER BY, LIMIT, Filtering	Q1, Q2, Q3
Level 2 — Aggregation	COUNT, SUM, GROUP BY, HAVING	Q4, Q5, Q6
Level 3 — Joins	INNER JOIN, multi-table JOIN patterns	Q7, Q8, Q9
Level 4 — Profit Analytics	Arithmetic expressions, 4-table JOINS	Q10, Q11
Level 5 — Customer Intelligence	Subqueries, nested SELECT, HAVING	Q12, Q13
Level 6 — Strategic BI	MONTH(), date functions, revenue ranking	Q14, Q15, Q16
Capstone — Full Simulation	All of the above combined in P&L reports	5 Reports

--- End of Assignment ---

Submitted by: Chetan Prajapat

[ACE_Project_Cheta](#)[nPraja](#)[pat](#)